well-structured lab exercises focused on:

- ✅ DOM Traversal
- ✅ Event Delegation
- ✅ Dynamic DOM Manipulation
- ✅ Medium → Complex Level

These are classroom-ready assignments.

🧪 LAB 1 – DOM Traversal Playground (Medium)

🎯 Objective

Practice:

- parentNode
- children
- firstElementChild
- lastElementChild
- nextElementSibling
- previousElementSibling

🧱 Starter HTML

```
<div id="container">
  <h2>Student List</h2>
  <ul id="students">
    <li>John</li>
    <li>David</li>
    <li>Emma</li>
    <li>Olivia</li>
```

```
  </ul>
</div>

<button id="highlightFirst">Highlight First</button>
<button id="highlightLast">Highlight Last</button>
<button id="highlightNext">Highlight Next of David</button>
```

📝 **Tasks**

1. When clicking **Highlight First**
   - Change first student text color to blue.
2. When clicking **Highlight Last**
   - Add background color to last student.
3. When clicking **Highlight Next of David**
   - Find David using traversal.
   - Highlight next sibling.
4. Print parent of <ul> in console.

💡 **Expected Skills**

- DOM selection
- Traversal navigation
- Styling using .style

🧪 **LAB 2 – Clickable Table with Traversal (Medium+)**

🎯 **Objective**

Use parentNode and children to extract row data.

🧱 Starter HTML

```
<table border="1" id="studentTable">
<tr>
  <th>Name</th>
  <th>Marks</th>
</tr>
<tr>
  <td>John</td>
  <td>85</td>
</tr>
<tr>
  <td>Emma</td>
  <td>92</td>
</tr>
</table>
```

📝 Tasks

1. When clicking any <td>:
   - Highlight entire row.
   - Print student name and marks in console.
2. Use traversal only (no extra selectors).

Hint:

- event.target.parentNode
- row.children

🧪 LAB 3 – Dynamic List with Event Delegation (Important)

🎯 Objective

Implement event delegation for dynamic content.

🧱 Starter HTML

```
<input type="text" id="taskInput">

<button id="addBtn">Add Task</button>

<ul id="taskList"></ul>
```

📝 Tasks

1. Add tasks dynamically.
2. When clicking a task:
    - Toggle line-through.
3. Double-click:
    - Remove task.
4. Use only ONE event listener on <ul>.

💡 Required Concepts

- event.target
- tagName
- appendChild
- remove()

🧪 LAB 4 – Nested Menu Traversal (Medium → Complex)

🎯 Objective

Handle nested elements with traversal.

🧱 **Starter HTML**

```
<ul id="menu">
  <li>Electronics
    <ul>
      <li>Mobile</li>
      <li>Laptop</li>
    </ul>
  </li>
  <li>Clothing
    <ul>
      <li>Men</li>
      <li>Women</li>
    </ul>
  </li>
</ul>
```

📝 **Tasks**

1. When clicking main category:
   - Toggle visibility of child <ul>.
2. Use:
   - children
   - nextElementSibling
3. Add smooth transition (CSS optional).

🧪 **LAB 5 – Advanced Event Delegation Dashboard (Complex)**

🎯 Objective

Build interactive UI with single event handler.

🧱 Starter Layout

```
<div id="dashboard">
  <button data-action="add">Add Box</button>
  <button data-action="remove">Remove Last</button>
  <button data-action="color">Change Color</button>

  <div id="boxContainer"></div>
</div>
```

📝 Tasks

1. Use ONE event listener on #dashboard.
2. Based on data-action:
    - Add a box
    - Remove last box
    - Change all box colors randomly
3. Boxes should be clickable to toggle border.

💡 Must Use

- event.target.dataset.action
- Event delegation
- Traversal
- createElement
- appendChild

🧪 LAB 6 – Complex Mini Project: Shopping Cart UI

🎯 Objective

Combine:

- DOM traversal
- Event delegation
- Dynamic manipulation

Requirements

1. Display product list (name + price + Add button)
2. When clicking Add:
    - Add to cart section
3. Clicking cart item:
    - Remove from cart
4. Show total price dynamically
5. Use event delegation for:
    - Product buttons
    - Cart items